The growth-law correction constant $\delta = \log R_\infty$ of the quadratic polynomial continued fraction: structure, a closed-form bracket, and certified high-precision values

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Abstract

For integers $A, B, C \geq 1$ let $b_k = Ak^2 + Bk + C$ and let q_n be the standard continuant denominators of the polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n\geq 1} 1/(An^2 + Bn + C)$ ($q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$). A companion note proves the growth law $\log q_n = n \log A + 2 \log(n!) + \frac{B}{A} \log n + (K_\Gamma + \delta) + \kappa/n + O(1/n^2)$, in which the product constant $K_\Gamma = -\log(\Gamma(1 - r_1)\Gamma(1 - r_2))$ is closed-form but the correction $\delta = \log R_\infty > 0$, with $R_\infty = \lim_n q_n / \prod_{k\leq n} b_k$, is left undetermined. This note *characterizes* δ. We show that the scaled limit R_∞ is exactly the *independence polynomial of a path graph* evaluated at the weights $u_n = 1/(b_{n-1}b_n)$, giving a cluster expansion $R_\infty = \sum_{k\geq 0} \sigma_k$ with $\sigma_k \leq S^k$, $S = \sum_{n\geq 2} u_n$. From it we derive a closed-form two-sided bracket $\log(1 + S) \leq \delta \leq -\log(1 - S)$, with S in closed digamma form (generic poles) and confluent polygamma form (degenerate poles; e.g. $S(1, 2, 1) = \pi^2/3 - 13/4$ exactly). Using the first three cluster sums $\sigma_1, \sigma_2, \sigma_3$ in exact closed form (machine-verified against brute-force enumeration to $\sim 10^{-54}$) to seed an exact downward recurrence, we evaluate δ to certified high precision: $\delta(1, 0, 1)$ to ≥ 44 digits, with a nine-triple table, each value certified two internal ways and cross-checked by a fully independent forward/Neville extrapolation. An integer-relation (PSLQ/`identify`) search returns no low-height closed form, and a precision-stability test shows the uncontrolled relations to be numerical artifacts; we record δ as conjecturally a new transcendental constant of the family. Finally, the finitary structural core — the Casoratian identity, the monotone–bounded convergence $R_n \to R_\infty$, and the closed-form bracket $1 + S \leq R_\infty \leq 1/(1 - S)$ — is machine-checked in Lean 4/Mathlib with clean axiom cones (no `sorry`).

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1 Introduction

The convergent denominators of the quadratic polynomial continued fraction $V(A, B, C) = 1 + \mathbf{K}_{n\geq 1} 1/(An^2 + Bn + C)$, integers $A, B, C \geq 1$, obey $q_0 = 1$, $q_1 = b_1$, $q_n = b_n q_{n-1} + q_{n-2}$ with $b_k = Ak^2 + Bk + C$. A companion note [1] establishes the exact two-sided growth law

$$\log q_n = n \log A + 2 \log(n!) + \tfrac{B}{A} \log n + \big(K_\Gamma + \delta\big) + \frac{\kappa}{n} + O(1/n^2), \tag{1}$$

where r_1, r_2 are the roots of $Ax^2 + Bx + C$, $K_\Gamma = -\log(\Gamma(1-r_1)\Gamma(1-r_2))$ is the *product constant* (at $B = 0$ equal to $\log(\sinh(\pi\sqrt{C/A})/(\pi\sqrt{C/A}))$), $\kappa = \frac{1}{2}(B^2/A^2 + B/A - 2C/A)$, and

$$\delta = \log R_\infty > 0, \qquad R_\infty = \lim_{n\to\infty} \frac{q_n}{\prod_{k=1}^n b_k} \tag{2}$$

is the *correction constant* produced by the $+q_{n-2}$ term of the recurrence. The product piece K_Γ is closed-form; the companion note claims *no* closed form for δ and treats it as a black box. This note opens the box.

Our contributions are, in increasing order of analytic depth:

(i) **Structure.** R_∞ is exactly the independence polynomial of the path graph on $\{2, \ldots, n\}$ in the limit, evaluated at $u_n = 1/(b_{n-1}b_n)$ (Proposition 1); equivalently a Mayer-type cluster expansion $R_\infty = \sum_k \sigma_k$ with $\sigma_k \le S^k$.

(ii) **A closed-form bracket.** $\log(1+S) \le \delta \le -\log(1-S)$ (Proposition 2), where $S = \sum_{n\ge2} u_n$ is closed-form via a digamma residue sum, with a confluent polygamma form at degenerate poles; one degenerate value is exactly rational-plus-π^2: $S(1,2,1) = \pi^2/3 - 13/4$.

(iii) **Certified high precision.** Exact closed forms for $\sigma_1, \sigma_2, \sigma_3$ seed an exact downward recurrence, yielding δ to ≥ 44 digits for $(1,0,1)$ and a nine-triple table (Section 5); each value is certified by seed-order independence and self-convergence, and cross-checked by an independent extrapolation method.

(iv) **Non-elementarity (conjectural).** An integer-relation search finds no low-height closed form, and a precision-stability test exhibits the uncontrolled relations as artifacts (Section 6).

(v) **A machine-checked core.** The finitary structural facts on which (i)–(iii) rest — the Casoratian identity, the monotone–bounded convergence $R_n \to R_\infty$, and the closed-form bracket $1 + S \le R_\infty \le 1/(1-S)$ (conditional on $\sum u_n < \infty$ and $S < 1$) — are formalized in Lean 4/Mathlib with clean axiom cones (Section 7).

Relation to prior work. The denominators q_n satisfy the three-term recurrence $q_n = b_n q_{n-1} + q_{n-2}$ with polynomially growing coefficients $b_n = An^2 + Bn + C \to \infty$, a Poincaré-type difference equation. Such a recurrence has a two-dimensional solution space spanned by a *dominant* solution (here q_n) and a *recessive* (minimal) solution; convergence of the continued fraction to V is exactly the selection of the minimal ray by p_n/q_n, the content of Pincherle's theorem (originally due to Pincherle; modern statement and attribution in [3], with minimal/recessive solutions in [7, §3.6]). The leading large-n asymptotics of growing-coefficient recurrences are governed by Perron's theorem [4] and the Birkhoff–Trjitzinsky/Wong–Li asymptotic theory [5, 6]; applied to q_n these produce precisely the dominant terms $n \log A + 2\log(n!) + \frac{B}{A}\log n$ and the Gamma-product constant K_Γ of the companion law (1). What that theory leaves undetermined is the additive *constant* — the amplitude relating the recessive solution to the product scale $\prod_k b_k$. The present note isolates and characterizes that constant, $\delta = \log R_\infty$: we show R_∞ is the path independence polynomial at the weights $u_n = 1/(b_{n-1}b_n)$ (Proposition 1), from which the closed-form bracket and the certified values below follow. To our knowledge neither this combinatorial identity for the amplitude nor the digamma closed form for its bracketing series S appears in that literature.

Standing assumptions. Throughout A, B, C are integers ≥ 1, so $b_k = Ak^2 + Bk + C \ge A + B + C \ge 3 > 0$ for all $k \ge 1$, $q_n > 0$, and $P_n := \prod_{k=1}^n b_k > 0$. The roots satisfy $r_1 + r_2 = -B/A < 0$, $r_1 r_2 = C/A > 0$, hence either form a complex-conjugate pair or are two negative reals; in all cases $\operatorname{Re} r_i < 0$.

Epistemic convention. We use the four-class SIARC partition. **PROVEN** = machine-checked in Lean with a clean axiom cone (a subset of {`propext`, `Classical.choice`, `Quot.sound`} and no `sorryAx`); **STRUCTURAL** = complete elementary hand proof, not yet formalized; **VERIFIED** = confirmed numerically for the stated finite cases; **CONJECTURED** = supported by evidence, not established. Each numbered result below is tagged accordingly, and Section 8 collects the grading. All numerical claims are reproduced by the scripts named in Section 9.

2 The scaled recurrence and the Casoratian frame

Let $R_n := q_n/P_n$, so $R_0 = R_1 = 1$. The convergent numerators p_n ($p_{-1} = 1, p_0 = 1$) satisfy the same second-order recurrence; set $P_n^{\mathrm{num}} := p_n/P_n$.

Lemma 1 (Scaled recurrence; STRUCTURAL, Lean core in Section 7)**.** *For all $n \geq 2$,*

$$R_n = R_{n-1} + u_n R_{n-2}, \qquad u_n := \frac{1}{b_{n-1}b_n} > 0.$$

Consequently (R_n) is strictly increasing, bounded above by $\prod_{k\geq 2}(1+u_k) < \infty$, and converges to a limit $R_\infty \in (1,\infty)$ with $R_\infty - R_n = O(1/n^3)$; thus $\delta = \log R_\infty > 0$.

Proof. Divide $q_n = b_n q_{n-1} + q_{n-2}$ by $P_n = b_n P_{n-1} = b_n b_{n-1} P_{n-2}$ to get $R_n = R_{n-1} + R_{n-2}/(b_{n-1}b_n)$. Since $u_n > 0$ and $R_0, R_1 > 0$, induction gives $R_n > 0$ and the increment $R_n - R_{n-1} = u_n R_{n-2} > 0$, so (R_n) strictly increases; monotonicity gives $R_{n-2} \leq R_{n-1}$, whence $R_n \leq R_{n-1}(1+u_n)$ and $R_n \leq \prod_{k=2}^{n}(1+u_k)$. As $b_{k-1}b_k \geq A^2(k-1)^2k^2$, the series $\sum_k u_k = O(\sum k^{-4})$ converges, so the product converges and (R_n) is bounded; a monotone bounded sequence converges to $R_\infty \geq R_2 = 1+u_2 > 1$. The tail estimate $R_\infty - R_n = \sum_{m>n} u_m R_{m-2} \leq R_\infty \sum_{m>n} u_m = O(1/n^3)$ follows by comparison with $\int_n^\infty t^{-4}\,dt$. □

Lemma 2 (Casoratian; PROVEN, Section 7)**.** *The discrete Wronskian of p_n, q_n is $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$; hence, in scaled variables,*

$$P_n^{\mathrm{num}} R_{n-1} - P_{n-1}^{\mathrm{num}} R_n = \frac{(-1)^{n-1}}{b_n\big(\prod_{k=1}^{n-1} b_k\big)^2}.$$

The initial data $p_{-1}{=}1$, $p_0{=}1$, $q_{-1}{=}0$, $q_0{=}1$ give $p_1{=}b_1{+}1$ and $q_1{=}b_1$, so Lemma 2 is exactly the convergent Casoratian $p_{n+1}q_n - p_n q_{n+1} = (-1)^n$ of Theorem 1 re-indexed by $n \mapsto n+1$; the Lean statement uses that shifted index and the equivalent sign $(-1)^n$.

The Wronskian decays like $P_n^{-2} \asymp (n!)^{-4}$, which is why the *value* $V = \lim p_n/q_n$ converges super-geometrically (matched to 77–117 digits across the test triples already at $N \approx 4000$, and to the 129 digits used in Section 6 at larger N) while $R_\infty - R_n$ decays only like $1/n^3$. The slow convergence of R_n is what forces the dedicated high-precision method of Section 5; a naive forward recurrence stalls at $\sim$ 25–38 digits.

3 R_∞ as an independence polynomial

The recurrence $R_n = R_{n-1} + u_n R_{n-2}$ is *not* the classical continuant recurrence $K_n = x_n K_{n-1} + K_{n-2}$: the weight u_n multiplies the *second-order* term. It is instead the transfer recurrence of an independence polynomial.

Proposition 1 (Independence-polynomial identity; VERIFIED exactly, STRUCTURAL)**.** *For every n,*

$$R_n = \sum_{\substack{T\subseteq\{2,\dots,n\}\\ \text{no two consecutive}}} \prod_{i\in T} u_i,$$

the independence polynomial of the path graph on the vertices $\{2,\dots,n\}$ (edges between consecutive integers), evaluated at vertex weights u_i. Letting $n\to\infty$,

$$R_\infty = \sum_{k\ge 0}\sigma_k, \qquad \sigma_0 = 1, \quad \sigma_1 = S = \sum_{n\ge 2} u_n, \quad \sigma_k = \sum_{\substack{T\subseteq\{2,3,\dots\},\ |T|=k\\ \text{gaps}\ge 2}} \prod_{i\in T} u_i. \tag{3}$$

Because each non-consecutive k-subset is one of the unrestricted k-tuples (which double-count order, repeats, and adjacency), $0\le\sigma_k\le S^k$, so the series converges geometrically.

Proof. Both sides satisfy the same recurrence and initial data. The empty set gives the constant term 1 ($R_1 = 1$); a non-consecutive subset $T\subseteq\{2,\dots,n\}$ either omits n (contributing to R_{n-1}) or contains n, in which case it omits $n-1$ and the rest is a non-consecutive subset of $\{2,\dots,n-2\}$ with the factor u_n (contributing u_nR_{n-2}). This is exactly $R_n = R_{n-1} + u_nR_{n-2}$, and the $n = 1, 2$ base cases match. The bound $\sigma_k\le S^k$ is the stated over-count. □

The cluster identity (3) is verified *exactly* (subset dynamic-programming equals R_n for all tested triples and all n) by the script `continuant_verify.py`.

4 A closed-form two-sided bracket

Proposition 2 (Bracket; PROVEN (Lean, conditional core; Section 7), VERIFIED all triples)**.** *With $S = \sum_{n\ge 2} u_n < 1$,*

$$1 + S \le R_\infty \le \frac{1}{1-S} \qquad \Longleftrightarrow \qquad \log(1+S)\le\delta\le -\log(1-S).$$

Proof. From $R_\infty = 1 + \sum_{n\ge 2} u_nR_{n-2}$ and $R_{n-2}\ge 1$ comes the lower bound $R_\infty\ge 1+S$. For the upper bound, by (3) and $\sigma_k\le S^k$, $R_\infty = \sum_k\sigma_k \le \sum_k S^k = 1/(1-S)$ (using $S<1$, which holds since numerically $S\le S(1,0,1)\approx 0.1307$). Apply log. □

The bracket is deliberately coarse: the lower bound keeps only $\sigma_0+\sigma_1$ and the upper bound replaces each σ_k by its geometric majorant S^k. The exact cluster forms σ_2,σ_3 of Section 5 (Appendix A) sharpen it well past the bracket.

Remark 1 (Machine-checked, cluster-free route)**.** *The R_∞-form of the bracket is formalized in Lean as `rinf_bracket` (Section 7),* conditional *only on the elementary analytic input that the partial sums $S_n = \sum_{k=2}^n u_k$ are bounded above and $S = \sup_n S_n < 1$. The formalization does not use the cluster expansion* (3)*: it rests on the two finitary inequalities $1 + S_n \le R_n$ (`rseq_ge_one_add_psum`) and $\left(\prod_{k=2}^n(1+u_k)\right)(1-S_n)\le 1$ (`rmaj_mul_one_sub_psum_le_one`), the latter from the exact identity $(1-S_n)-(1+u)(1-S_n-u) = u\,S_n + u^2\ge 0$. The δ-form follows by monotonicity of* log.

It remains to put S in closed form. Write $b_n = A(n-r_1)(n-r_2)$, so $1/(b_{n-1}b_n)$ has its poles in the multiset $\{r_1, r_2, 1+r_1, 1+r_2\}$.

Table 1: The closed-form bracket $\log(1+S) \le \delta \le -\log(1-S)$ on representative triples; δ from the high-precision method of Section 5.

(A,B,C)	S	$\log(1+S)$	δ	$-\log(1-S)$
$(1,0,1)$	0.13066961899	0.12281004	0.12385719436	0.14003204
$(1,0,2)$	0.080228466794	0.077172562	0.077741108481	0.083629973
$(2,0,1)$	0.045720089039	0.044705729	0.044814419813	0.046798243
$(1,1,1)$	0.065740306948	0.063669682	0.064033816183	0.068000835
$(3,2,5)$	0.0068561031151	0.0068327069	0.0068373312816	0.0068797142

Proposition 3 (Closed form for S; VERIFIED to 35–62 digits). *If the four poles are distinct,*

$$S = -\sum_{\rho} \mathrm{Res}_\rho\ \psi(2-\rho), \qquad \mathrm{Res}_\rho = \frac{1}{A^2 \prod_{\sigma \ne \rho}(\rho - \sigma)}, \tag{4}$$

with ψ the digamma function (the sum of residues vanishes, $\sum_\rho \mathrm{Res}_\rho = 0$, reflecting $1/(b_{n-1}b_n) = O(n^{-4})$, so the individually divergent ψ-pieces combine convergently). For repeated poles, partial fractions over the multiset of multiplicities m_ρ give the confluent form

$$S = -\sum_{\rho} a_{\rho,1}\, \psi(2-\rho) + \sum_{\rho} \sum_{j \ge 2} a_{\rho,j}\, \frac{(-1)^j\, \psi^{(j-1)}(2-\rho)}{(j-1)!}, \tag{5}$$

$a_{\rho,j}$ the Laurent coefficients of $1/(b_{n-1}b_n)$ at ρ, with $\sum_\rho a_{\rho,1} = 0$.

Equations (4)–(5) are the termwise sums $\sum_{n\ge 2}(n-\rho)^{-1}$ (regularized, $j=1$) and $\sum_{n \ge 2}(n-\rho)^{-j} = (-1)^j \psi^{(j-1)}(2-\rho)/(j-1)!$ ($j \ge 2$), standard polygamma series [7, §5.15]. Both are confirmed against direct summation (`mpmath.nsum`) for distinct- and repeated-pole triples (`S_closed_form.py`, `S_confluent.py`). A clean degenerate value: for $(A,B,C) = (1,2,1)$, $b_n = (n+1)^2$ has poles $\{-1,-1,0,0\}$ and

$$S(1,2,1) = \frac{\pi^2}{3} - \frac{13}{4} = 0.0398681336964528729\ldots, \tag{6}$$

matched to 10^{-40}. (Under the standing assumptions $A, B, C \ge 1$ every $b_k = Ak^2 + Bk + C \ge 3 > 0$, so no summation index is a pole and the residue and confluent forms apply throughout.) Table 1 shows the bracket on representative triples.

5 Certified high-precision values

Because $R_n \to R_\infty$ only like $1/n^3$, the forward recurrence is useless beyond a few dozen digits. We instead run the *exact downward recurrence*

$$T(m) = T(m+1) + u_m\, T(m+2), \qquad T(\infty) = 1, \qquad R_\infty = T(2), \tag{7}$$

where $T(m)$ is the independence polynomial of the tail $\{m, m+1, \dots\}$ (so $T(m) = \sum_{k \ge 0} \sigma_k(m)$ with $\sigma_k(m)$ the order-k cluster sum starting at m). The recurrence (7) is exact, so *all* error lives in the seed $T(M+1), T(M+2)$ at the truncation index M. Truncating the cluster series at order

Table 2: Certified high-precision $\delta(A, B, C) = \log R_\infty$ via the exact order-3 cluster seed and downward recurrence (7) at $M = 10^5$ (working precision 90 digits), each value self-checked against $M = 2.5 \times 10^4$ and cross-checked by independent forward/Neville extrapolation. "rel. dig." is the conservatively certified number of correct digits.

(A, B, C)	$\delta = \log R_\infty$ (first 30 digits)	rel. dig.
$(1, 0, 1)$	0.123857194360626392728504989702	$44^\dagger$
$(1, 0, 2)$	0.077741108480647487187139168954	39
$(1, 0, 3)$	0.055023822866781361284024549668	39
$(2, 0, 1)$	0.044814419812755566106936562111	40
$(3, 0, 1)$	0.023024249675539505266856933463	41
$(1, 1, 1)$	0.064033816182909523022426151229	39
$(1, 2, 1)$	0.039254537227423694454156302525	39
$(2, 1, 3)$	0.017942479040119357169479600823	40
$(3, 2, 5)$	0.006837331281558730523182815470	41

† certified to ≥ 44 digits at $M = 10^6$, dps$=$ 170; see (10).

p leaves seed error $\sim \sigma_{p+1} = O(M^{-3(p+1)})$, i.e. about $3(p+1)$ correct digits per decade of M. We use $p = 3$ (~ 12 digits/decade), for which exact closed forms are available:

$$\sigma_2 = \tfrac{1}{2}\big(p_1^2 - p_2\big) - a_1, \tag{8}$$

$$\sigma_3 = \tfrac{1}{6}\big(p_1^3 - 3p_1p_2 + 2p_3\big) - a_1p_1 + c + t, \tag{9}$$

with power sums $p_j = \sum_{n \geq m} u_n^j$ (and $p_1 = \sigma_1(m) = S(m)$ from Proposition 3, exact via ψ), and adjacency sums $a_1 = \sum_n u_n u_{n+1}$, $c = \sum_n u_n u_{n+1}(u_n + u_{n+1})$, $t = \sum_n u_n u_{n+1} u_{n+2}$.

Proposition 4 (Cluster closed forms; STRUCTURAL (Appendix A), VERIFIED to $\sim 10^{-54}$)**.** *Equations* (8)–(9) *hold: the unrestricted elementary symmetric sums* $e_2 = \frac{1}{2}(p_1^2 - p_2)$, $e_3 = \frac{1}{6}(p_1^3 - 3p_1p_2 + 2p_3)$ *corrected for adjacency by inclusion–exclusion (Appendix A). They agree with the brute-force enumeration of non-consecutive* 2- *and* 3-*subsets for* (A, B, C) *equal to* $(1, 0, 1)$, $(1, 1, 1)$, $(2, 1, 3)$, $(1, 2, 1)$ *to better than* 10^{-40} *(residuals* $\sim 10^{-54}$–10^{-55}*).*

This is the gate of `verify_clusters.py`: no derived combinatorial identity is trusted until machine-checked against brute force. With the gate passed, the downward recurrence (7) seeded by $1 + \sigma_1(M) + \sigma_2(M) + \sigma_3(M)$ gives the values in Table 2.

For the reference triple $(1, 0, 1)$ we certify δ *three* independent ways at $M = 10^6$ (working precision 170 digits): seed-order independence (order-2 vs order-3 seed agree to 56 digits; order-3 vs order-3-plus-leading-σ_4 to 75 digits), self-convergence ($M = 10^5$ vs 10^6 agree to 47 digits), and a fully independent forward continuant with Neville extrapolation in $1/N$ (agrees to ~ 25 digits, its own resolution ceiling). The conservative reliable count is ≥ 44 digits:

$$\delta(1, 0, 1) = 0.12385719436062639272850498970259084096757955\ldots \tag{10}$$

Two robustness notes. The degenerate triple $(1, 2, 1)$ has coincident poles, so the simple-pole residue form (4) is singular there; the seed's $\sigma_1(m) = S(m)$ must use the exact confluent polygamma form (5) (a direct numerical tail sum loses precision at large m). With that substitution $(1, 2, 1)$ matches the others (39 digits), and its $S(2)$ reproduces (6) exactly. Separately, the independent forward/Neville cross-check must start the scaled continuant at the index that includes the u_2 term, or it converges to a different limit; with the correct start it agrees with the cluster value to ~ 25 digits across all tested triples.

On the previously reported "80-digit" figure. An earlier internal record cited an 80-digit value for $\delta(1,0,1)$. The present method reaches 80 digits in principle at $M = 10^7$ (consistent with the ~ 12 digit/decade rate), but we *certify* only the ≥ 44 digits of (10) at feasible cost; nothing is asserted beyond the certified count.

6 No low-height closed form for δ

Proposition 5 (Non-elementarity; CONJECTURED, rigorous null). *No low-height integer relation is found for V, R_∞, or δ against natural constant bases. Specifically:* `identify`(V) *= None at* 100 *digits (the strongest null, V being the natural "answer", known to* 129 *digits via the fast p_n/q_n ratio);* `identify`(δ) *= None; PSLQ over $\{1, \pi, \log 2, \gamma, \text{Catalan}, \zeta(3)\}$ with* $\max |c_i| \leq 200$ *returns no relation; and R_∞ — the natural algebraic candidate, tested by PSLQ on $\{1, R_\infty, R_\infty^2, R_\infty^3, R_\infty^4\}$ — is not algebraic of degree ≤ 4 and height $\leq 10^4$ (δ and V receive the* `identify` *null above).*

Uncontrolled PSLQ does return high-height vectors, but `pslq_stability.py` shows them to be artifacts: the recovered vector wanders with working precision ($[30, 110, \dots] \to [-244, \dots] \to [1330, \dots] \to [-25296, \dots]$) and the residual floors at $\sim 10^{-\text{dps}}$ rather than shrinking toward true zero — whereas a genuine integer relation is precision-invariant. We therefore record δ as conjecturally a genuine new transcendental constant of the quadratic family, not reducible to the product constant K_Γ. Irrationality of δ is open. A null integer-relation result is, we stress, a complete and honest answer, not a gap.

7 The machine-checked finitary core

The finitary facts underlying Sections 2–4 are formalized in Lean 4 with Mathlib (toolchain pinned: `lean4:v4.30.0`, Mathlib `rev=v4.30.0`). Each declaration was checked with `#print axioms`; every axiom cone is a subset of the Mathlib defaults {`propext`, `Classical.choice`, `Quot.sound`} with *no* `sorryAx`, the source contains no `sorry`/`admit`, and the build is warning- and error-free (exit 0).

Theorem 1 (Formalized core; PROVEN). *The following are machine-checked with clean axiom cones:*

- `casoratian_step`, `casoratian_eq`, `pq_casoratian`: *the Casoratian of two solutions of $s_{n+2} = b_{n+2}s_{n+1} + s_n$ flips sign each step, hence equals $(-1)^n$ times its initial value; for the convergents ($p_0{=}1, p_1{=}b_1{+}1, q_0{=}1, q_1{=}b_1$) this is $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$ (over an arbitrary* `CommRing`*).*

- `rseq_ge_one`, `rseq_mono`, `rseq_monotone`, `rmaj_ge_one`, `rmaj_mono`, `rseq_le_rmaj`: *the scaled sequence $r_{n+2} = r_{n+1} + u_{n+2}r_n$ ($r_0{=}r_1{=}1$, $u \geq 0$) satisfies $r_n \geq 1$, is monotone, and is dominated by the partial products $\prod_{k=2}^{n}(1 + u_k)$ (over $\mathbb{R}$).*

- `rseq_bddAbove_of_rmaj`, `rseq_tendsto_ciSup`, `rseq_tendsto_of_rmaj_bddAbove`: *monotone convergence — if the partial products are bounded above (equivalently $\sum u_n < \infty$) then $R_n \to \sup_n R_n$, i.e. R_∞ exists.*

- *`psum`, `psum_nonneg`, `rseq_ge_one_add_psum`, `rmaj_mul_one_sub_psum_le_one`, `rinf_bracket`, `rseq_tendsto_under_psum_bdd`: the* closed-form bracket $1 + S \le R_\infty \le 1/(1 - S)$ *(Proposition 2), conditional on the partial sums $S_n = \sum_{k=2}^{n} u_k$ being bounded above and $S = \sup_n S_n < 1$; the finitary core is $1 + S_n \le R_n$ and $\big(\prod_{k=2}^{n}(1 + u_k)\big)(1 - S_n) \le 1$ (over $\mathbb{R}$).*

This is the *conditional-core* pattern: the finitary algebra and order structure are proved unconditionally, while the single analytic input — that the partial products are bounded above — is taken as an explicit labelled hypothesis (it is the convergence of $\sum u_n = O(\sum k^{-4})$, elementary but analytic). Out of scope for the formalization, and proved by hand/numerics above, are: the *value* of R_∞, the closed forms for S, the product constant K_Γ and the Stirling/Gamma asymptotics of (1), and the non-elementarity conjecture.

8 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 1): the Casoratian identity, the existence of $R_\infty = \lim R_n$ via monotone–bounded convergence, and the closed-form bracket $1 + S \le R_\infty \le 1/(1-S)$ (Proposition 2), the last two conditional on the elementary analytic input $\sum u_n < \infty$ (and $S < 1$ for the bracket).

- **STRUCTURAL** (complete elementary hand proof; not yet formalized): the scaled recurrence and tail estimate (Lemma 1), the independence-polynomial identity (Proposition 1), and the closed forms for S and σ_2, σ_3 (Propositions 3, 4; the inclusion–exclusion derivation of σ_2, σ_3 is Appendix A).

- **VERIFIED** (numerically): the cluster identity exactly (`continuant_verify.py`); σ_2, σ_3 and each step of their inclusion–exclusion derivation to $\sim 10^{-54}$ (`verify_clusters.py`, `verify_inclusion_exclusion.py`); S closed forms to 35–62 digits vs direct summation; the bracket on all tabulated triples; and the δ values of Table 2, certified ≥ 39 digits each and ≥ 44 for $(1, 0, 1)$, with an independent cross-check.

- **CONJECTURED**: δ has no low-height elementary closed form (Proposition 5); δ is a new transcendental constant of the family. Irrationality of δ is open.

9 Code and data availability

Every numerical claim is reproduced by the following scripts (mpmath): the harness `harness_rinf.py` (two-solution frame, R_∞, value V); `continuant_verify.py` (exact independence-polynomial check); `S_closed_form.py`, `S_confluent.py` (closed forms for S); `verify_clusters.py` and `verify_inclusion_exclusion` (the σ_2, σ_3 gate and the step-by-step inclusion–exclusion check, Proposition 4, Appendix A); `hp_delta_v2.py` and `finalize_m2.py` (the high-precision δ values and Table 2, output `delta_values_m2.txt`); `verify_forward_crosscheck.py` (the independent cross-check); `hp_probe.py`, `pslq_stability.py` (the integer-relation nulls, Proposition 5); and the Lean package `lean/pcf_continuant` (`Basic.lean`, `Bracket.lean`, `Check.lean`; Theorem 1). Version 1.0 of this note is archived at Zenodo (version DOI `10.5281/zenodo.20578401`, concept DOI `10.5281/zenodo.20578400`) as a supplement to the companion growth-law note [1]; the present version adds the machine-checked closed-form bracket (`rinf_bracket`, Theorem 1). The Lean sources and the verification scripts named above accompany the deposit in its public repository, `https://github.com/papanokechi/pcf-delta`.

A Inclusion–exclusion derivation of the cluster closed forms

This appendix proves Equations (8)–(9) by elementary inclusion–exclusion, upgrading Proposition 4 from a numerically VERIFIED identity to a complete STRUCTURAL hand proof; the brute-force gate of `verify_clusters.py` and the step-by-step gate of `verify_inclusion_exclusion.py` then confirm every line to $\sim 10^{-54}$.

Fix the tail $\{m, m+1, \dots\}$ with vertex weights $u_i > 0$ and power sums $p_j = \sum_{i \geq m} u_i^j$. Recall from (3) that $\sigma_k = \sum \prod_{i \in T} u_i$ runs over k-subsets T with no two consecutive indices, and let $e_k = \sum_{|T|=k} \prod_{i \in T} u_i$ be the *unrestricted* elementary symmetric sum. Newton's identities give

$$e_2 = \tfrac{1}{2}\big(p_1^2 - p_2\big), \qquad e_3 = \tfrac{1}{6}\big(p_1^3 - 3p_1p_2 + 2p_3\big). \tag{11}$$

Derivation of (8). Each unrestricted pair $\{i < j\}$ is either non-consecutive (counted by σ_2) or consecutive ($j = i+1$); the consecutive pairs sum to $a_1 = \sum_i u_i u_{i+1}$. Hence $\sigma_2 = e_2 - a_1 = \frac{1}{2}(p_1^2 - p_2) - a_1$. □

Derivation of (9). Classify the unrestricted triples $\{i < j < k\}$ by their number of consecutive pairs. Let $\mathcal{B}$ be those with at least one, so $\sigma_3 = e_3 - \Sigma(\mathcal{B})$, where Σ denotes the weighted sum $\sum \prod_{i \in T} u_i$. A triple has either *exactly one* consecutive pair — an adjacent pair plus an isolated third vertex; call this set $\mathcal{B}_1$ — or *two* — a run $\{i, i+1, i+2\}$; call this $\mathcal{B}_2$ — and $\Sigma(\mathcal{B}_2) = \sum_i u_i u_{i+1} u_{i+2} = t$.

Expand the product of the adjacency sum and the first power sum:

$$a_1 p_1 = \Big(\sum_i u_i u_{i+1}\Big)\Big(\sum_r u_r\Big) = \sum_i \sum_r u_i u_{i+1} u_r.$$

Split the inner sum according to whether $r \in \{i, i+1\}$. The terms with $r = i$ or $r = i+1$ sum to $\sum_i u_i u_{i+1}(u_i + u_{i+1}) = c$. The remaining terms, with $r \notin \{i, i+1\}$, each form a triple — the adjacent pair $\{i, i+1\}$ together with a distinct third vertex r — whose total we call W, so that $a_1 p_1 = c + W$. In W each member of $\mathcal{B}_1$ is counted once (its unique adjacent pair fixes $(i, i+1)$ and r is the isolated vertex), while each run $\{a, a+1, a+2\}$ in $\mathcal{B}_2$ is counted twice (as the pair $(a, a+1)$ with $r = a+2$, and as the pair $(a+1, a+2)$ with $r = a$). Thus $W = \Sigma(\mathcal{B}_1) + 2t$, and

$$\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t = (W - 2t) + t = W - t = a_1 p_1 - c - t.$$

Subtracting from e_3 in (11),

$$\sigma_3 = e_3 - \Sigma(\mathcal{B}) = \tfrac{1}{6}\big(p_1^3 - 3p_1p_2 + 2p_3\big) - a_1 p_1 + c + t. \qquad \square$$

The script `verify_inclusion_exclusion.py` checks each labelled step independently against brute force over a finite window (e_2, e_3 against the elementary sums; a_1 against the adjacent pairs; $a_1 p_1 = c + W$; $W = \Sigma(\mathcal{B}_1) + 2t$; $\Sigma(\mathcal{B}) = \Sigma(\mathcal{B}_1) + t$; and the two closed forms), for $(A, B, C) \in \{(1,0,1), (1,1,1), (2,1,3), (1,2,1)\}$, all to $\sim 10^{-54}$.

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